Foundations of Universal approximation theorem

[Universal approximation theorem]

$$f(x) = \sum_{i=1}^{N} v_i \sigma(w_i^T x + b_i)$$

1.0 Content Block

Reservoir computing and quantum reservoir computing In reservoir computing a sparse recurrent neural network with fixed weights equipped of fading memory and echo state property is followed by a trainable output layer. Its universality has been demonstrated separately for what concerns networks of rate neurons and spiking neurons, respectively. In 2024, the framework has been generalized and extended to quantum reservoirs where the reservoir is based on qubits defined over Hilbert spaces.

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Arbitrary width The first results concerned the arbitrary width case. Ken-ichi Funahashi (May 1989) showed that Rumelhart–Hinton–Williams type backpropagation networks possess universal approximation capability with a class of sigmoidal activation functions, extending the result to multi-output mappings as well. Kurt Hornik, Maxwell Stinchcombe, and Halbert White (July 1989) showed that multilayer feed-forward networks with as few as one hidden layer are universal approximators, provided that the activation function satisfies certain conditions. George Cybenko (December 1989) independently established a related result for sigmoid activation functions using functional-analytic methods. Hornik also showed in 1991 that it is not the specific choice of the activation function but rather the multilayer feed-forward architecture itself that gives neural networks the potential of being universal approximators. Moshe Leshno et al in 1993 and later Allan Pinkus in 1999 showed that the universal approximation property is equivalent to having a nonpolynomial activation function.

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Arbitrary-depth case The "dual" versions of the theorem consider networks of bounded width and arbitrary depth. A variant of the universal approximation theorem was proved for the arbitrary depth case by Zhou Lu et al. in 2017. They showed that networks of width n + 4 with ReLU activation functions can approximate any Lebesgue-integrable function on n-dimensional input space with respect to L 1 {\displaystyle L^{1}} distance if network depth is allowed to grow. It was also shown that if the width was less than or equal to n, this general expressive power to approximate any Lebesgue integrable function was lost. In the same paper it was shown that ReLU networks with width n + 1 were sufficient to approximate any continuous function of n-dimensional input variables. The following refinement, specifies the optimal minimum width for which such an approximation is possible and is due to.

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This is an existence result. It says that activation functions providing universal approximation property for bounded depth bounded width networks exist. Using certain algorithmic and computer programming techniques, Guliyev and Ismailov efficiently constructed such activation functions depending on a numerical parameter. The developed algorithm allows one to compute the activation functions at any point of the real axis instantly. For the algorithm and the corresponding computer code see. The theoretical result can be formulated as follows.

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Bounded depth and bounded width case The first result on approximation capabilities of neural networks with bounded number of layers, each containing a limited number of artificial neurons was obtained by Maiorov and Pinkus. Their remarkable result revealed that such networks can be universal approximators and for achieving this property two hidden layers are enough.

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Arbitrary depth The arbitrary depth case was also studied by a number of authors such as Gustaf Gripenberg in 2003, Dmitry Yarotsky, Zhou Lu et al in 2017, Boris Hanin and Mark Sellke in 2018 who focused on neural networks with ReLU activation function. In 2020, Patrick Kidger and Terry Lyons extended those results to neural networks with general activation functions such, e.g. tanh or GeLU. One special case of arbitrary depth is that each composition component comes from a finite set of mappings. In 2024, Cai constructed a finite set of mappings, named a vocabulary, such that any continuous function can be approximated by compositing a sequence from the vocabulary. This is similar to the concept of compositionality in linguistics, which is the idea that a finite vocabulary of basic elements can be combined via grammar to express an infinite range of meanings.

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Setup Artificial neural networks are combinations of multiple simple mathematical functions that implement more complicated functions from (typically) real-valued vectors to real-valued vectors. The spaces of multivariate functions that can be implemented by a network are determined by the structure of the network, the set of simple functions, and its multiplicative parameters. A great deal of theoretical work has gone into characterizing these function spaces. Most universal approximation theorems are in one of two classes. The first quantifies the approximation capabilities of neural networks with an arbitrary number of artificial neurons ("arbitrary width" case) and the second focuses on the case with an arbitrary number of hidden layers, each containing a limited number of artificial neurons ("arbitrary depth" case). In addition to these two classes, there are also universal approximation theorems for neural networks with bounded number of hidden layers and a limited number of neurons in each layer ("bounded depth and bounded width" case).

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Here " σ : R → R {\displaystyle \sigma \colon \mathbb {R} \to \mathbb {R} } is λ {\displaystyle \lambda } -strictly increasing on some set X {\displaystyle X} " means that there exists a strictly increasing function u : X → R {\displaystyle u\colon X\to \mathbb {R} } such that | σ (x) − u (x) | ≤ λ {\displaystyle |\sigma (x)-u(x)|\leq \lambda } for all x ∈ X {\displaystyle x\in X} . Clearly, a λ {\displaystyle \lambda } -increasing function behaves like a usual increasing function as λ {\displaystyle \lambda } gets small. In the "depth-width" terminology, the above theorem says that for certain activation functions depth- 2 {\displaystyle 2} width- 2 {\displaystyle 2} networks are universal approximators for univariate functions and depth- 3 {\displaystyle 3} width- (2 d + 2) {\displaystyle (2d+2)} networks are universal approximators for d {\displaystyle d} -variable functions (d > 1 {\displaystyle d>1}).

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The above proof has not specified how one might use a ramp function to approximate arbitrary functions in C 0 (R n , R) {\displaystyle C_{0}(\mathbb {R} ^{n},\mathbb {R})} . A sketch of the proof is that one can first construct flat bump functions, intersect them to obtain spherical bump functions that approximate the Dirac delta function, then use those to approximate arbitrary functions in C 0 (R n , R) {\displaystyle C_{0}(\mathbb {R} ^{n},\mathbb {R})} . The original proofs, such as the one by Cybenko, use methods from functional analysis, including the Hahn-Banach and Riesz–Markov–Kakutani representation theorems. Cybenko first published the theorem in a technical report in 1988, then as a paper in 1989. Notice also that the neural network is only required to approximate within a compact set K {\displaystyle K} . The proof does not describe how the function would be extrapolated outside of the region. The problem with polynomials may be removed by allowing the outputs of the hidden layers to be multiplied together (the "pi-sigma networks"), yielding the generalization: